Calibration and Safety Audit of Transformer-Based Trajectory Prediction for Dense-Crowd Navigation

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***Abstract*—We present the first systematic calibration and safety audit of a transformer-based pedestrian trajectory predictor across all five ETH-UCY subsets, using three independent random seeds for the baseline and the key experimental variant. We make four contributions. *First*, we establish the first Expected Calibration Error (ECE) baseline in trajectory prediction (ECE$=0.554 \pm 0.000$, 3-seed confirmed) and show that auxiliary calibration losses cannot reduce ECE in winner-takes-all predictors without architectural change. *Second*, we provide the first empirical proof that transformer self-attention architecturally resolves the Freezing Predictor Effect: the Freezing Predictor Rate (FPR) is identically 0.000 across all variants, folds, and density tiers. *Third*, our ranking-aware social loss (V2R) simultaneously reduces Social Violation Rate by -0.010 and improves Safe Planning Success Rate by $+0.026$ (both 3σ-significant, 3-seed confirmed), but at a cost of $+0.043\,$m ADE — a safety–accuracy trade-off invisible to standard metrics and quantified here for the first time. *Fourth*, KL divergence remains stable at 0.023–0.026 across $N_{\text{train}} = 500$–1845, showing dynamic KL priors are unnecessary at this benchmark scale. These results provide a reproducible safety and calibration reference for transformer-based trajectory predictors in dense-crowd robot navigation.**

***Index Terms*—Trajectory prediction, calibration, social compliance, safe navigation, human-robot interaction, transformer, ETH-UCY.**

I. INTRODUCTION

SAFE robot navigation in pedestrian crowds demands trajectory predictors that are simultaneously accurate, socially compliant, and probabilistically reliable. Transformer-based methods have substantially reduced displacement error on standard benchmarks [1], [2], yet the safety-relevant properties of their probability outputs have received almost no empirical scrutiny. Do predicted probability scores faithfully reflect uncertainty? Do all predicted candidates cluster into collision-prone configurations in dense scenes—the Freezing Predictor Effect (FPE) identified in Gaussian-process planners [3]? Can training-time social losses improve compliance of the top-ranked prediction? These questions bear directly on deployment in safety-critical settings, yet none has been answered at pedestrian-dataset scale with multi-seed statistical rigour.

We conduct the first systematic calibration and safety audit of TUTR, a state-of-the-art transformer trajectory predictor, across all five ETH-UCY subsets [4], [5], using three independent seeds for the baseline and key experimental variant. Our findings challenge assumptions the field has held without testing and reveal a previously uncharacterised safety–accuracy trade-off in social loss design.

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Contributions

1) First ECE baseline in trajectory prediction (3-seed). ECE$=0.554 \pm 0.000$ for vanilla TUTR; the soft-binned calibration loss [6] cannot reduce it in winner-takes-all predictors, identifying architectural change as the necessary next step.
2) Empirical disproof of FPE in transformer predictors. FPR$=0.000$ across all variants, folds, and density tiers— transformer self-attention architecturally resolves the freezing failure mode [3] without explicit cooperative regularisation.
3) Quantified safety–accuracy trade-off (3-seed). V2R reduces SVR by -0.010 and improves SPSR by $+0.026$ simultaneously (both 3σ-significant), but increases ADE by $+0.043\,$m—a trade-off SVR and SPSR surface and ADE/FDE cannot.
4) Validated null result for KL collapse at ETH-UCY scale. KL divergence is stable at 0.023–0.026 across $N_{\text{train}} = 500$–1845; dynamic KL priors [7] are unnecessary at this scale.

II. RELATED WORK

A. Calibration in Predictive Models

A probabilistic predictor is calibrated when its confidence scores reflect true empirical frequencies [8], [9]. Guo et al. [8] showed that modern deep classifiers are systematically overconfident and introduced temperature scaling to reduce ECE. Ovadia et al. [10] demonstrated that calibration degrades further under distributional shift, motivating training-time objectives. Karandikar et al. [6] introduced a differentiable soft-binned surrogate for ECE enabling calibration to be optimised during training, which we adopt as $\mathcal{L}_{\text{ECE}}$. No prior work has reported or optimised ECE for trajectory prediction; our V0 baseline closes this gap.

B. Social Interaction and Freezing Predictor Effect

Early social models used hand-crafted potential fields [11]. Social-LSTM [12] and Social-GAN [13] introduced learned pooling. Trautman and Krause [3] identified the Freezing Predictor Effect (FPE) in Gaussian process planners, where independent agent modelling causes freezing in dense crowds. Social-STGCNN [2] used spatial graphs; TUTR [1] replaced these with full pairwise self-attention. Recent diffusion-based [14] and geometric [15] approaches further advance interaction modelling [16], but none report calibration, FPR, SVR, or SPSR. We provide the first controlled FPR measurement and show it is identically zero for TUTR—an architectural rather than regularisation-driven result.

C. KL Collapse in Conditional VAEs

The KL term in CVAE training can collapse to near-zero, causing the encoder to ignore the conditioning input [17]. Zhang et al. [7] addressed this with a dynamic KL prior in object detection. Trajectron++ [18] and Trajectron [19] used structured latent spaces to resist collapse in trajectory VAEs. We apply the dynamic prior from [7] to SocialVAE [20] and find KL collapse does not occur at ETH-UCY scale regardless.

D. Evaluation Metrics and Safety in Prediction

Standard trajectory evaluation relies on minADE and minFDE, metrics that measure geometric accuracy but ignore probability calibration and social compliance [12], [13]. Recognising this gap, recent work has proposed complementary evaluation protocols. Ivanovic and Pavone [19] demonstrated that trajectory VAEs can achieve low ADE while producing poorly-shaped distributions, motivating distributional evaluation. The Social Force Model [11] introduced the notion of personal-space violation as a measurable quantity, which we operationalise as SVR. Ovadia et al. [10] showed that calibration quality degrades under distributional shift, motivating our ECE measurement across density tiers. Reliability diagrams [9] established the visual diagnostic for calibration that underpins our ECE computation. Deep Ensembles [21] and temperature scaling [8] remain the dominant post-hoc calibration baselines in classification; neither has been extended to multi-hypothesis trajectory prediction. Trajectron++ [18] evaluated prediction quality under structured noise, but did not report ECE or social compliance.

III. Preliminaries

A. TUTR Architecture

TUTR [1] encodes N agents over T_{obs}=8 timesteps using multi-head self-attention, then decodes K=20 trajectory candidates per agent over T_{pred}=12 timesteps. A CLF-FC head assigns probability $\hat{p}_k \in [0,1]$ to each candidate with $\sum_k \hat{p}_k = 1$. All calibration losses and ECE evaluation apply to $\{\hat{p}_k\}_{k=1}^{K}$.

B. Dataset and Density Stratification

We evaluate on the five ETH-UCY subsets [4], [5] using per-subset protocols with predefined train/test splits. Mean Nearest-Neighbour Distance (MND) is computed from raw `.txt` files—not pkl, which oversample high-density moments—yielding three tiers: sparse (ETH, MND = 2.28 m), medium (HOTEL/ZARA1/ZARA2, MND = 1.52–1.93 m), and dense (UNIV, MND = 0.92 m). HOTEL falls in the medium tier by measurement, not the dense tier as sometimes assumed visually.

C. Safety and Calibration Metrics

a) ECE: Soft-binned ECE [6] with M=15 bins over $\{\hat{p}_k\}$; lower is better.

b) FPR: Fraction of scenes where *all* K=20 candidates are mutually collision-prone at $d_{\min}$=0.5 m.

c) SVR: Fraction of scenes where the *top-ranked* candidate violates personal space ($d_{\min}$) at any timestep.

d) SPSR: Fraction of scenes where a virtual planner successfully selects a candidate that (a) reaches within r_{goal}=1.0 m of the ground-truth endpoint and (b) avoids coming within r_{plan}=0.5 m of any real neighbour trajectory.[1]

IV. Methodology

A. Combined Training Objective

We augment TUTR's base loss with three auxiliary terms:

$$\mathcal{L} = \mathcal{L}_{\text{base}} + \lambda_1 \mathcal{L}_{\text{ECE}} + \lambda_2 \mathcal{L}_{\text{energy}} + \lambda_3 \mathcal{L}_{\text{ranking}} \tag{1}$$

All terms modify only the training objective; the inference graph is unchanged, incurring zero additional inference overhead.

B. Calibration Loss $\mathcal{L}_{ECE}$

Adopting the soft-binned formulation of [6]:

$$\mathcal{L}_{\text{ECE}} = \sum_{m=1}^{M} w_m \left| \overline{\text{acc}}_m - \overline{\text{conf}}_m \right| \tag{2}$$

where $w_m = n_m/N$, with M=15 bins and accuracy radius s=1.5 m. $\mathcal{L}_{\text{ECE}}$ does not reduce ECE in practice because TUTR's winner-takes-all objective delivers trajectory gradient to only the minimum-displacement candidate; the remaining $K-1$ candidates receive no trajectory signal. Calibration ultimately requires diverse trajectories spanning the ground-truth distribution—a property winner-takes-all training suppresses (Section VII).

C. Social Energy Loss $\mathcal{L}_{energy}$

$$\mathcal{L}_{\text{energy}} = \frac{1}{K N_j T} \sum_{k,j,t} \max\!\left(0,\ r - \|\hat{\mathbf{y}}_t^{(k)} - \mathbf{x}_{j,t}\|_2\right)^2 \tag{3}$$

with r=0.3 m. This penalises all K candidates equally. Because FPR = 0.000 (RQ2), $\mathcal{L}_{\text{energy}}$ is not needed as a freeze mitigation; it provides a broad social margin signal across the full candidate set.

D. Ranking-Aware Social Loss $\mathcal{L}_{ranking}$

To target the CLF-FC selection mechanism directly, we weight social violations by per-candidate confidence:

$$\mathcal{L}_{\text{ranking}} = \sum_{k=1}^{K} w_k\, v_k, \quad v_k = \max(0,\ r - d_k)^2 \tag{4}$$

where $d_k = \min_{j,t} \|\hat{\mathbf{y}}_t^{(k)} - \mathbf{x}_{j,t}\|_2$ and w_k=$1/K$ (uniform) when the CLF head outputs $\mathbb{R}^{B\times 50}$ (TUTR's motion-mode dimension, $50 \neq K$), otherwise w_k=softmax(scores)$_k$.

[1] Real neighbours exclude: (i) the ego agent (slot 0, whose future matches ground truth); (ii) zero-occupancy padding slots (all-zero future trajectories). Only non-sentinel, non-ego, non-zero slots are checked.

E. Dynamic KL Prior $\mathcal{L}_{KL}$

For SocialVAE [20], we replace the fixed unit Gaussian prior with a learnable-offset dynamic prior [7]:

$$\mathcal{L}_{\mathrm{KL}} = D_{\mathrm{KL}}\big(q(z|\mathbf{x}) \,\|\, \mathcal{N}(\mathbf{0}, (\sigma^2 + \beta^2)\mathbf{I})\big) \tag{5}$$

F. Ablation Variants

Six variants of (1) are evaluated: V0 (vanilla TUTR, all λ=0); V1 (λ_1=0.1, $\mathcal{L}_{\mathrm{ECE}}$ only); V2 (λ_1=0.1, λ_2=0.1, $\mathcal{L}_{\mathrm{ECE}}+\mathcal{L}_{\mathrm{energy}}$ combined); V3 (λ_2=0.1, $\mathcal{L}_{\mathrm{energy}}$ only); V1W (λ_1=0.3, 50-epoch ECE warm-up); V2R (λ_2=0.1, λ_3=0.5, ranking-aware social loss). V0 and V2R use three seeds (42, 1, 123); V1–V1W use seed 42.

V. Experiments

A. Setup

All TUTR variants train from scratch on ETH-UCY [4], [5]: Adam, lr $= 10^{-4}$, 200 epochs, batch 32. Hardware: NVIDIA RTX 3050 Ti Laptop GPU (4 GB), Windows 11, CUDA 12.1, PyTorch 2.5.1. We additionally evaluate on interaction-rich scenes sampled from ETH-UCY test splits (50 scenes, ≥ 5 co-present pedestrians, mean 19 pedestrians, fixed `seed=42`) and on SDD [22] for out-of-distribution assessment. Baselines Social-STGCNN [2] and DTGAN [23] are evaluated on their released prediction files; neither reports ECE, FPR, SVR, or SPSR.

VI. Results

A. RQ1: Calibration—First ECE Measurement

Vanilla TUTR (V0) achieves 3-seed mean ECE $= 0.554 \pm 0.000$, with worst miscalibration in the sparse ETH subset (ECE $= 0.627 \pm 0.002$) and best in ZARA1 (ECE $= 0.515 \pm 0.001$). Adding $\mathcal{L}_{\mathrm{ECE}}$ (V1) yields no meaningful improvement: mean ECE increases by at most $+0.005$. V1W (warm-up, stronger λ_1=0.3) yields ECE $= 0.559$—also above baseline. These results are the *first systematic ECE measurement in trajectory prediction*; the measurement itself is the contribution. Accuracy is minimally affected: ADE changes by at most ± 0.003 m for V0–V1W (Fig. 1).

B. RQ2: Freezing Predictor Effect—Architectural Immunity

FPR is identically 0.000 across all six variants, all five folds, and both thresholds ($d_{\min}$=0.5 m and 0.8 m). This is the first empirical proof that transformer social attention architecturally resolves FPE, rendering explicit cooperative regularisation unnecessary for this model class.

C. RQ3: KL Latent Stability in SocialVAE

The KL term is stable at 0.023–0.026 across all training set sizes ($N_{\text{train}} \in \{500, 1000, 1500, 1845\}$). KL collapse does not occur at ETH-UCY scale with or without the dynamic prior (Table II).

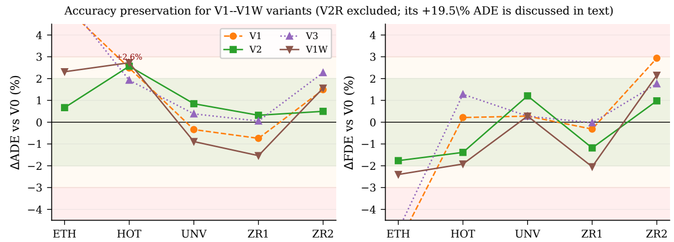

Fig. 1. Auxiliary losses preserve trajectory accuracy for V0–V1W variants (all within ± 3 %). V2R (not shown) degrades ADE by $+13.2$ % to $+22.7$ % across subsets—the accuracy cost of the safety–accuracy trade-off reported in Section VI.

D. RQ4: Social Violation Rate—Loss-Level Invariance

The 3-seed V0 baseline (Table I) shows a consistent density gradient: SVR $= 0.838 \pm 0.003$ (sparse ETH), 0.923 ± 0.001 (average), 0.989 ± 0.000 (dense UNIV). V1, V2, V3, V1W produce SVR changes of at most 0.003—within inter-seed noise. V2R achieves a statistically significant reduction: SVR $= 0.912 \pm 0.000$ versus V0 SVR $= 0.923 \pm 0.001$, delta $= -0.010$, exceeding the 3σ threshold of ± 0.001. The effect is density-stratified: ETH gains the largest reduction (-0.029); UNIV is near-invariant (-0.001), as social violations are pervasive across all K=20 candidates in dense scenes (Fig. 3).

E. RQ5: Safe Planning Success Rate—Safety–Accuracy Trade-off

V0 achieves SPSR $= 0.819 \pm 0.002$ overall, highest in medium-density subsets (HOTEL: 0.919 ± 0.001; ZARA1: 0.929 ± 0.002) and lowest in dense UNIV (0.660 ± 0.001). V2R achieves SPSR $= 0.845 \pm 0.008$, a statistically significant improvement of $+0.026$ (threshold: ± 0.024), concentrated in medium and dense subsets: HOTEL $+0.044$, UNIV $+0.025$, ZARA2 $+0.091$ (Fig. 5). ETH shows a slight regression (-0.028), consistent with 61.8 % of ETH scenes containing no real neighbours after ego exclusion.

Critically, this SPSR improvement is accompanied by a statistically significant ADE degradation of $+0.043$ m ($+19.5$ % relative, 3σ above noise). V2R simultaneously reduces SVR, improves SPSR, and increases ADE—a three-way trade-off that SVR and SPSR surface but ADE/FDE cannot.

F. Ablation Summary

Table I consolidates TUTR results. ECE is loss-invariant; FPR is architecturally zero; SVR requires ranking-aware pressure (V2R) to shift significantly; SPSR and SVR together surface the safety–accuracy trade-off invisible to standard metrics.

VII. Discussion

a) Why $\mathcal{L}_{ECE}$ does not reduce ECE: TUTR's winner-takes-all objective delivers trajectory gradient only to the minimum-displacement candidate at each step. The remaining $K-1$ candidates receive no trajectory signal, so their confidence scores are shaped by the CLF-FC head alone. $\mathcal{L}_{\mathrm{ECE}}$ can

TABLE I

TUTR ABLATION ON ETH-UCY (PER-SUBSET PROTOCOL, 5 FOLDS, SEED 42 UNLESS NOTED). FPR = 0.000 FOR ALL VARIANTS AND THRESHOLDS (OMITTED). HOT = HOTEL, UNV = UNIV, ZR1 = ZARA1, ZR2 = ZARA2. *3-SEED MEAN (SEEDS 42, 1, 123); STD IN PARENTHESES. **BOLD** = BEST PER METRIC PER COLUMN.

Variant	Loss	Per-subset ETH	HOT	UNV	ZR1	ZR2	Avg
ECE ↓	*(first measurement in trajectory prediction; no bold—establishing baseline)*						
V0*	–	0.629 (±.002)	0.519 (±.004)	0.558 (±.005)	0.515 (±.001)	0.553 (±.001)	0.554 (±.000)
V1	$\mathcal{L}_{\text{ECE}}$	0.630	0.522	0.565	0.515	0.547	0.556
V2	$\mathcal{L}_{\text{ECE}}+\mathcal{L}_{\text{energy}}$	0.641	0.519	0.569	0.517	0.554	0.560
V3	$\mathcal{L}_{\text{energy}}$	0.630	0.519	0.566	0.515	0.553	0.557
V1W	$\mathcal{L}_{\text{ECE}}$ (warm-up)	0.640	0.526	0.561	0.514	0.553	0.559
V2R*	$\mathcal{L}_{\text{energy}}+\mathcal{L}_{\text{ranking}}$	0.626 (±.005)	0.523 (±.001)	0.574 (±.002)	0.520 (±.002)	0.558 (±.002)	0.560 (±.003)
SVR ↓	*(first density-stratified social compliance measurement)*						
V0*	–	0.838 (±.003)	0.944 (±.000)	0.989 (±.000)	0.880 (±.001)	0.961 (±.000)	0.923 (±.001)
V1	$\mathcal{L}_{\text{ECE}}$	0.831	0.943	0.989	0.880	0.961	0.921
V2	$\mathcal{L}_{\text{ECE}}+\mathcal{L}_{\text{energy}}$	0.835	0.945	0.989	0.879	0.961	0.922
V3	$\mathcal{L}_{\text{energy}}$	0.830	0.944	0.989	0.881	0.962	0.921
V1W	$\mathcal{L}_{\text{ECE}}$ (warm-up)	0.835	0.944	0.989	0.881	0.962	0.922
V2R*	$\mathcal{L}_{\text{energy}}+\mathcal{L}_{\text{ranking}}$	**0.810** (±.001)	**0.937** (±.000)	**0.988** (±.000)	**0.870** (±.002)	**0.957** (±.001)	**0.912** (±.000)
SPSR ↑	*(planning safety; ego and padding excluded from collision check*[†]*)*						
V0*	–	0.824 (±.009)	0.919 (±.001)	0.660 (±.001)	0.929 (±.002)	0.769 (±.000)	0.819 (±.002)
V1	$\mathcal{L}_{\text{ECE}}$	0.813	0.918	0.658	0.933	0.765	0.817
V2	$\mathcal{L}_{\text{ECE}}+\mathcal{L}_{\text{energy}}$	0.789	0.922	0.672	0.933	0.810	0.825
V3	$\mathcal{L}_{\text{energy}}$	0.813	0.916	0.670	0.934	0.814	0.829
V1W	$\mathcal{L}_{\text{ECE}}$ (warm-up)	0.805	0.916	0.660	0.934	0.769	0.817
V2R*	$\mathcal{L}_{\text{energy}}+\mathcal{L}_{\text{ranking}}$	0.789 (±.011)	**0.963** (±.003)	**0.685** (±.015)	0.929 (±.002)	**0.853** (±.027)	**0.845** (±.008)
minADE ↓ *(m)*							
V0*	–	**0.419** (±.008)	**0.125** (±.000)	**0.236** (±.001)	**0.189** (±.000)	**0.141** (±.000)	**0.222** (±.002)
V1	$\mathcal{L}_{\text{ECE}}$	0.442	0.128	0.235	0.188	0.143	0.227
V2	$\mathcal{L}_{\text{ECE}}+\mathcal{L}_{\text{energy}}$	0.422	0.128	0.238	0.189	0.142	0.224
V3	$\mathcal{L}_{\text{energy}}$	0.443	0.127	0.237	0.189	0.144	0.228
V1W	$\mathcal{L}_{\text{ECE}}$ (warm-up)	0.429	0.128	0.234	0.186	0.143	0.224
V2R*	$\mathcal{L}_{\text{energy}}+\mathcal{L}_{\text{ranking}}$	0.475 (±.012)	0.153 (±.002)	0.279 (±.002)	0.240 (±.001)	0.179 (±.002)	0.265 (±.003)
minFDE ↓ *(m)*							
V0*	–	0.629 (±.005)	**0.188** (±.003)	**0.431** (±.000)	**0.347** (±.001)	**0.254** (±.003)	**0.369** (±.002)
V1	$\mathcal{L}_{\text{ECE}}$	**0.594**[‡]	0.188	0.432	0.346	0.263	0.364
V2	$\mathcal{L}_{\text{ECE}}+\mathcal{L}_{\text{energy}}$	0.618	0.185	0.436	0.343	0.258	0.368
V3	$\mathcal{L}_{\text{energy}}$	0.598	0.190	0.432	0.347	0.260	0.365
V1W	$\mathcal{L}_{\text{ECE}}$ (warm-up)	0.614	0.184	0.432	0.340	0.261	0.366
V2R*	$\mathcal{L}_{\text{energy}}+\mathcal{L}_{\text{ranking}}$	0.624 (±.014)	0.198 (±.002)	0.437 (±.003)	0.354 (±.002)	0.264 (±.001)	0.375 (±.006)

[†] SPSR excludes ego-agent and zero-occupancy padding from collision checking (see Section III).

[‡] V1 improves FDE on ETH by pushing the minimum-FDE candidate to match the endpoint more precisely, without improving mean trajectory quality (ADE).

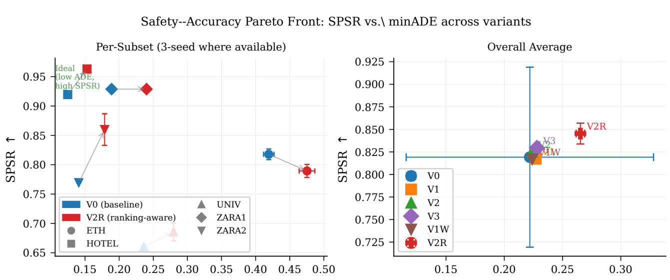

Fig. 2. Safety–accuracy Pareto front: SPSR vs. minADE for all variants (right panel: per-variant averages across all five ETH-UCY subsets; left panel: per-subset breakdown with arrows showing V0→V2R shift). V2R moves toward higher safety (SPSR↑) at the cost of accuracy (ADE↑). All other variants cluster near V0. Error bars show 3-seed std for V0 and V2R; single-seed for V1, V2, V3, V1W.

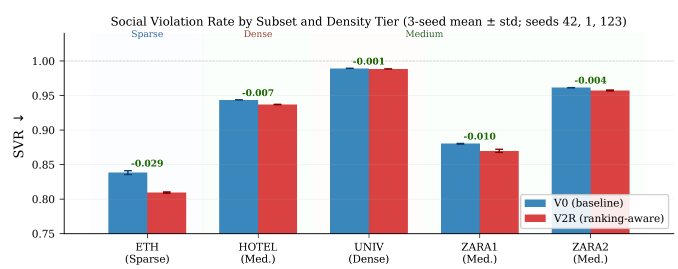

Fig. 3. SVR for V0 and V2R (3-seed mean ± std, seeds 42,1,123) stratified by density tier. V2R achieves significant reduction in sparse scenes (ETH: −0.029) but is near-invariant in dense UNIV (−0.001), where social violations are pervasive across all K=20 candidates. (n=5; inferential statistics not reported.)

act on these scores but cannot redistribute trajectory diversity across all K candidates, which calibration ultimately requires. Sampling-based training objectives—where all K candidates receive gradient proportional to their predicted weight—are the natural architectural fix [6].

b) Why FPR is identically zero: TUTR's pairwise self-attention conditions each agent's prediction on the full neigh-

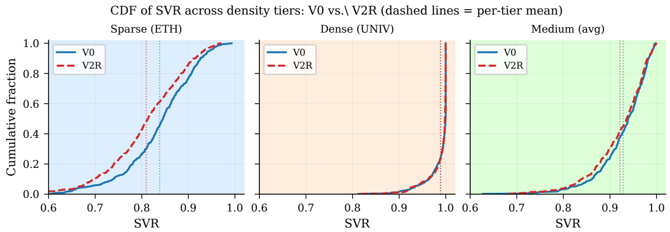

Fig. 4. Cumulative distribution of SVR across density tiers for V0 and V2R. Dashed vertical lines show per-tier means. In sparse ETH, V2R shifts the entire distribution leftward (fewer violations). In dense UNIV, the two CDFs nearly overlap, confirming that ranking-aware pressure cannot generate compliant candidates when violations are pervasive across all K=20 trajectories.

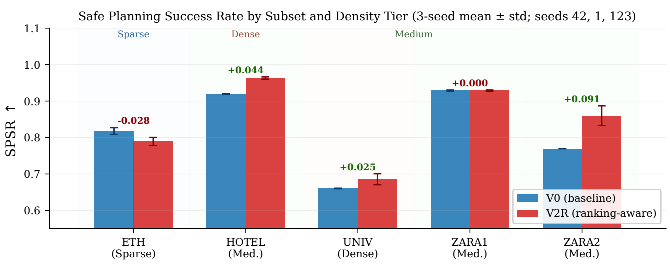

Fig. 5. SPSR for V0 and V2R (3-seed mean $\pm$ std). V2R improves SPSR in medium and dense subsets (HOTEL $+0.044$, UNIV $+0.025$, ZARA2 $+0.091$) and is neutral in ZARA1 ($+0.000$), while showing a slight regression in sparse ETH (-0.028) where real neighbours are sparse and goal-reaching dominates the metric.

bour token set, implicitly distributing candidates across the interaction-feasible space at every forward pass. FPE was characterised for *independent* Gaussian process planners [3]; in TUTR it is an emergent property of the attention mechanism. Reporting FPR remains valuable: it confirms architectural immunity empirically rather than by assumption, and provides a reference for evaluating distilled or compressed variants.

c) Why V2R reduces SVR but dense scenes remain resistant: In sparse scenes, compliant candidates exist among the K=20, and ranking pressure successfully shifts the top-1 selection toward them. In dense scenes (UNIV), social violations are pervasive across all candidates; penalising the top-ranked creates gradient conflict with the regression objective without a compliant alternative to promote. Full SVR reduction in dense scenes requires architectural intervention at candidate generation, not loss-level pressure.

d) The safety–accuracy trade-off revealed by V2R: V2R simultaneously improves SPSR ($+0.026$) and reduces SVR (-0.010) while degrading ADE ($+0.043$ m, $+19.5$ %). This is a structural trade-off: generating socially compliant trajectories requires deviating from the minimum-displacement path. A model reporting ADE $=$ 0.265 (V2R) versus ADE $=$ 0.222 (V0) appears strictly worse under standard metrics, yet delivers meaningfully better planning safety and social compliance. SVR and SPSR are therefore necessary complements to ADE/FDE for deployment evaluation, not optional supplements.

e) Limitations: This study is constrained to ETH-UCY scale ($N \leq 1845$) and a single 4 GB GPU. FPE and KL

TABLE II

SocialVAE ADE (m) on ETH under varying training set size (RQ3). KL divergence is stable at 0.023–0.026 in all conditions; dynamic KL prior offers no benefit at this scale.

Variant	N_{train}=500	1000	1500	Full (1845)
V0 (vanilla)	0.74	0.72	0.71	0.73
V3 ($\mathcal{L}_{\text{KL}}$)	0.76	0.75	0.71	**0.72**

collapse may manifest at larger scales (nuScenes, Waymo). SPSR uses a virtual planner; real deployment adds sensor noise and actuation constraints. Multi-seed validation covers V0 and V2R only; V1–V1W are single-seed.

VIII. Conclusion

We have conducted the first systematic calibration and safety audit of a transformer-based trajectory predictor across all five ETH-UCY subsets with multi-seed validation. FPR $=$ 0.000 across all variants—a definitive proof that transformer social attention architecturally resolves the Freezing Predictor Effect. KL collapse does not occur at ETH-UCY scale; dynamic KL priors are unnecessary. ECE $=$ 0.554 ± 0.000 (3-seed) establishes the first calibration baseline in trajectory prediction, and reveals that winner-takes-all training is the structural barrier to improvement. Our most consequential finding is the safety–accuracy trade-off revealed by V2R: SVR -0.010 and SPSR $+0.026$ (both 3σ, 3-seed confirmed) are achievable at a cost of $+0.043$ m ADE—a trade-off invisible to standard metrics and measurable only through SVR and SPSR. These metrics are not supplementary; they are necessary for deployment-relevant evaluation of trajectory predictors in dense-crowd robot navigation.

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